

Qikang Huang

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EDUCATION

University of Michigan, Ann Arbor

Sep. 2026 – May 2028

M.S. in Electrical and Computer Engineering

Track: Signal & Image Processing and Machine Learning

Communication University of China / Coventry University

Sep. 2022 – Jun. 2026

Sino-foreign Cooperative Education Program

B.Eng. in Intelligent Science and Technology (Communication University of China)

BSc (Hons) in Computer Science with Artificial Intelligence, First Class Honours (Coventry University)

- **GPA:** 88/100

- **Core Courses:** Large Models and Deep Learning (98), Neural Networks and Deep Learning (97), Natural Language Processing (89), Machine Learning (92), Data Structures (93), Computer Vision (91)

PROFESSIONAL EXPERIENCE

iFLYTEK Co., Ltd., Zhengzhou, China

Jan. 2026 – Apr. 2026

Assistant Research Algorithm Engineer – Smart Livestock Vision System

- Contributed to YOLOv8-based pig vision models for smart farming scenarios, including pig counting, posture detection, backfat-related classification, and abnormal-case monitoring.
- Refined the training-data feedback loop by validating 10,000+ RGB/depth images in LabelMe and correcting missed, false, duplicate, and inaccurate bounding-box annotations.
- Analyzed production failure cases from inspection robots and pig-counting modules, categorized false positives and false negatives, and supported model retraining and iteration.

ByteDance, Beijing, China

Jul. 2025 – Sept. 2025

Algorithm Intern – Loan Default Prediction Model Development and Optimization

- Built an end-to-end credit-risk prediction pipeline covering data cleaning, feature engineering, model training, hyperparameter tuning, and evaluation to predict whether users would default on loans.
- Engineered time, categorical, binned, correlation, time-window, and group-ratio features, and compared LightGBM, XGBoost, and CatBoost using stratified K-fold cross-validation and ROC-AUC.
- Applied PSI-based stability selection, L2 regularization, early stopping, and stacking; the optimized XGBoost model achieved an AUC of approximately 0.738.

Business Intelligence of Oriental Nations Corporation Ltd., Beijing, China

Jan. 2025 – Feb. 2025

Data Analyst Assistant – Telecom Traffic Revenue Forecasting

- Supported a telecom operator mobile data-traffic revenue forecasting project to predict future revenue and identify key business drivers.
- Analyzed traffic-revenue time series with ACF/PACF, built ARIMA, SARIMA, ETS, and Prophet models, and evaluated performance using MAE.
- Selected SARIMA as the best-performing model, completed a 14-month revenue forecast, and proposed optimization recommendations on traffic unit price, average usage, billed users, and zero-billed users.

RESEARCH EXPERIENCE

Retrieval Strategy Optimization for Multi-turn Legal Consultation Systems Based on RAG

Jan. 2026 – May 2026

Undergraduate Thesis

- Designed and implemented a RAG-based legal document retrieval optimization framework for multi-turn Chinese legal consultation, addressing context dependency, information omission, colloquial queries, and mismatch between user questions and formal legal expressions.
- Built an evaluation pipeline based on the LexRAG dataset, covering 17,228 Chinese legal articles and 5,065 multi-turn consultation queries, and systematically evaluated retrieval recall and ranking quality using Recall@K and NDCG@K.
- Integrated query rewriting, BGE-M3 hybrid retrieval, HyDE query enhancement, and Cross-Encoder reranking to improve candidate article recall and ranking performance.
- The HyDE-enhanced hybrid retrieval strategy improved Recall@10 from 0.3757 to 0.4701 and NDCG@10 from 0.2651 to 0.3349, providing empirical support for retrieval-chain optimization in legal RAG systems.
- Found that key bottlenecks were concentrated in query representation modeling and legal terminology expansion rather than simple reranking of candidate results.

SKILLS & INTERESTS

- **Technical Skills:** Vibe Coding, Python, PyTorch, NumPy, OpenCV, Matplotlib, LaTeX
- **Languages:** English (IELTS: 7.0, GRE: 325), Chinese
- **Interests:** Fitness, Hiking